Learning Object Recommendations for Teachers Based On Elicited ICT Competence Profiles

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Abstract—Recommender systems (RS) offer personalized services for facilitating the process of appropriate item selection. To perform this task, user profiling mechanisms should be implemented to automatically construct and update meaningful user profiles. These profiles can drive the RS in providing informed recommendations suited to the unique characteristics of each user. In the context of technology enhanced learning (TeL) Recommender Systems, the majority of research focus directly on learners' profiling and ignore the potential benefits of profiling teachers' professional capacities too. As a result, limited previous works exist on effectively capturing and utilizing individual teachers' particular professional characteristics, such as their Digital Competences (commonly referred to as ICT Competences) and exploiting these in systems that support their teaching preparation and practice, for example in the selection of appropriate educational resources. This paper proposes a RS which targets to support teachers in selecting learning objects (LO) from existing LO repositories (LORs) in a unified manner, namely by (a) automatically constructing their ICT Competence Profiles based on their actions within these LORs and (b) exploiting these profiles for more efficient LO selection. Experiments with data from three real-life LORs are presented and evaluation results are discussed to demonstrate the benefits of the proposed system.

Index Terms—Learning objects, personalized E-learning, web services, recommender systems

1 Introduction

Recommender systems (RS) are software tools designed to assist users in tackling the information overload problem by highlighting suitable items in a personalized manner [1]. They have been used in a wide range of application contexts, spanning from the movie industry [2] to e-commerce [3] and technology enhanced learning (TeL) [4].

RS have been implemented based on an increasing number of techniques, the most prominent and commonly used being the content-based filtering, the collaborative filtering, the demographic filtering and the hybrid approaches [1]. The content-based filtering systems generate recommendations based on users' past choices and the content similarity between the items that were favored in the past to novel ones that have not yet been discovered [5]. The collaborative filtering systems utilize the users' ratings over the available items for providing recommendations based on information provided by the "like-minded" neighbors of the active user (i.e., the user currently using the system and receiving the generated recommendations) [6]. The demographic filtering relies on the assummed commonalities that users with similar demographic backgrounds will have [7]. Finally, hybrid approaches combine techniques from the other approaches in order to reap the benefits of all, while tackling their individual drawbacks [8].

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Manuscript received 5 Jan. 2015; revised 20 Apr. 2015; accepted 2 May 2015. Date of publication 18 May 2015; date of current version 16 Mar. 2016. For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org, and reference the Digital Object Identifier below. Digital Object Identifier no. 10.1109/TLT.2015.2434824

In order to provide their personalized services, RS are usually implemented to automatically gather data from the users towards capturing their unique attributes, creating and updating individual profiles [9]. The reason for this is that users tend to either not manually provide personal data, or they do not provide them in an accurate manner, at a great cost to the RS prediction capacity in both occassions [9]. Therefore, and in-line with the general need for constructing and exploiting highly granulated user profiles in RS applications [10], intelligent mechanisms should be employed in order to collect and process meaningful user data for infering essential user characteristics that could provide added value in the quality of recommendations.

In the context of TeL, RS have been primarily utilized for the recommendation of different types of learning objects (LO) based on individual teachers' and learners' profiles. Nevertheless, although learners' profiling has been extensively considered, teachers' profiling is almost neglected [11]. Taking into consideration, the important role of teachers in adopting technology-supported school-based educational innovations [12], the complexity of technology-supported teaching practice [13] and the current diversity of individual teachers' digital competences [14], it is safe to assume that individual teachers' professional ICT competences profiling is an important element to be considered in educational RSs and, thus, it is worthy to investigate this topic.

In our previous work, we have proposed and evaluated a LO recommendation approach that aimed to support teachers in identifying and selecting educational resources for their course design and delivery, taking into consideration the individual teacher's current ICT competence status [15]. Furthermore, we have proposed and evaluated a mechanism for eliciting Fuzzy teachers' ICT Competence Profile (ICT-CP) based on their behavior patterns within learning object repositories (LOR) [16].

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Following these initial works, in this paper we propose and study a Recommender System that supports teachers, in a holistic manner, to select appropriate learning objects from existing LO Repositories (LORs) based on their dynamically updated ICT competence profile. The contribution of the proposed system is twofold. First, the RS utilizes the teachers' ICT Competence profiles, which are automatically elicited and dynamically updated based on the teachers' continuous usage behavior patterns within LORs. Second, the recommendation process itself adopts a new method for generating the LO recommendations by exploiting the previously mentioned elicited ICT Competence profiles, i.e., (a) by creating an ICT Competence-based neighborhood and (b) by promoting the LOs that the active teacher is currently competent in using.

The proposed system is presented and evaluated following a two-layer presentation in line with the commonly accepted layered evaluation methodology [17] of adaptive learning systems, which has been extended to RS in [18]. More specifically, a separate presentation and evaluation of the "*Teacher ICT Competence Profile Elicitation*" Layer and the "*Learning Object Recommendation Generation*" Layer is performed.

The remainder of this paper is structured as follows. Section 2 presents the background of the present study, as well as the identified research problem. Section 3 presents the proposed approach to teacher ICT Competence-based LO recommendations. This approach includes both a method for eliciting the teachers' ICT-CP based on their usage behavior within learning object repositories as well as the manner in which these profiles are exploited towards providing more informed ICT Competence-based LO recommendations, compared to existing, commonly used RS methods (described in Section 4). Section 4 presents a detailed analysis of the evaluation methodology employed. Section 5 presents the evaluation results for both Layers of the proposed RS. Finally, Section 6 contains the conclusions of the present work and the future work in this research agenda.

2 Background

2.1 User Profiling

User profiling is a technique that has been widely applied in a range of software applications, including adaptive web systems [19] and recommender systems [20]. It involves gathering data from the users in order to create a profile for each of them, depicting their unique attributes in the context of the system's application [21]. The latter could involve, for example, movie genre preferences in a movie recommender system [2], or learner preferences in an educational RS [22].

The process of creating individual user profiles in software applications is essential because each user has his/her own characteristics and needs. Therefore, capturing these user attributes is necessary for enabling the provision of personalized services [23]. A user profile, therefore, presents the system's full interpretation of the users' preferences and personal characteristics.

In many cases, user profiles are not provided by the users themselves but they are being elicited automatically through the identification, collection and processing of relevant user actions' data [24]. The reason for this is that users are usually either unwilling to provide such information or when they do, the validity and completeness of the provided data cannot be ensured [25]. Therefore, the elicitation process is usually based on the users' relevance feedback data [26].

Relevance feedback data can be attained in two ways, namely explicit feedback and implicit feedback [27]. The former requires users to perform specific actions that will inform and update their profile attributes, e.g., assign ratings to items or download items. This approach provides a set of benefits for the hosting RS system, such as increased development simplicity and enhanced accuracy in the profile update process [28]. Users' implicit feedback refers to mechanisms that monitor the users' interaction with the system in an unobtrusive manner. Such approaches have been developed in order to completely detach the user from the explicit feedback providing process and maximize the amount of data that are being harvested by the system [29]. Examples of user actions that are being monitored for profiling purposes include browsing time in each item and type of items accessed, uploaded or ignored [21].

2.2 User Profiling in Teacher-Oriented Recommender Systems

In the context of TeL, the majority of the implemented RS targets the learners and aims to provide them with personalized learning material and sequences of learning activities towards specific educational goal attainment [30]. The main *learner attributes* that are being used in such processes include their prior knowledge, learning preferences/styles, individual goals and other cognitive characteristics [31]. Subsequently, user profiling approaches have been primarily focused on accommodating the attributes of the learners [32].

However, despite the fact that learners are indeed the main focus of learning processes, other actors are also important for the successful implementation of effective learning procedures. More specifically, ***teachers*** also play a vital part in the educational processes and, amongst other reasons, their ICT competences [12], [14] and personal attitudes towards ICT use [33] can greatly affect the level and the quality of their technology-supported teaching practice. Therefore, systematic accommodation of teachers' professional characteristics should be an important design consideration for educational RS. Within this context, our previous work [11] included a literature review of existing teacher-oriented educational RS. The 22 identified approaches are summarized in Table 1. Table 1 also contains information on the types of relevance feedback data that are being harvested. These relevance feedback data include Explicit and Implicit data. The former include Social data (e.g., ratings and bookmarks) and user Demographic data, while the latter include, for example number of views of LO. Moreover, Table 1 presents information on whether these data are being utilized only for ad-hoc similarity calculations (i.e., they do not incorporate profile creation) or if they are being automatically processed and exploited to create dynamic and adaptive user profiles for providing personalized recommendations (i.e., they do incorporate profile creation).

Only (completed) systems utilizing the above techniques were considered. Table values designated as "no" either a lack of user profiling mechanism in the corresponding

TABLE 1
Existing Teacher-Oriented TeL Recommender Systems

	RS	Relevance Feedback Data	Profile Creation
1	[34]	Explicit: Demographic Explicit: Social Implicit	no
2	[35]	Explicit: Social	no
3	[36]	-	no
4	[37]	Explicit: Demographic Explicit: Social	no
5	[38]	Explicit: Social	no
6	[39]	Explicit: Social	no
7	[40]	Implicit	yes
8	[41]	Implicit Explicit: Social	yes
9	[42]	Explicit: Social	yes
10	[43]	-	no
11	[44]	-	no
12	[45]	Explicit: Social	no
13	[46]	Explicit: Social	no
14	[47]	Explicit: Social	no
15	[48]	-	no
16	[49]	Explicit: Social	no
17	[50]	Explicit: Social Implicit	yes
18	[51]	Explicit: Social	no
19	[52]	Explicit: Demographic Explicit: Social	yes
20	[53]	Explicit: Social	no
21	[54]	Explicit: Demographic Explicit: Social	no
22	[22]	Explicit: Social Implicit	yes

system or lack of specific presentation of relevant data in the presenting paper. As Table 1 depicts, existing teacher-oriented RS utilize a wide range of "raw" relevance feedback data types for creating user profiles. More specifically, most RS utilize "Explicit: Social" relevance feedback data ($N = 17$, $x = 77\%$). "Implicit" ($N = 5$, $x = 23\%$) and "Explicit: Demographic" ($N = 4$, $x = 18\%$) relevance feedback data are used less often. However, regardeless of the type of the relevance feedback utilized, the Table 1 data signify that only a small portion of existing teacher-oriented RS ($N = 6$, $x = 27\%$) create, maintain and exploit user profiles for providing enhanced recommendations to teachers, based on these relevance feedback data. This can be considered as potential drawback, since systematic and efficient user modeling is an essential element of successful RSs [10].

More specifically, Shelton et al. [40] describe a system that utilizes the teachers' clicks and time on each webpage to detect their preferences and alter the recommendations accordingly. Brusilovsky et al. [41] propose a social navigation approach for assisting teachers to identify useful educational resources within web-based repositories, through manipulation of their usage history and explicit feedback. Schoeffeger et al. [42] (and [50]) proposed a method for identifying emergent topics that teachers are dealing with and are interested in. Fazeli et al. [52] proposed a method for collecting both demographic and social data from teachers towards utilizing them for building trust networks. Finally, Ferreira-Satler et al. [22] propose an ontology-based fuzzy teacher profile inference system. The main idea is to elicit the preferences of the teachers based on the semantic importance of each LO that they create or browse. This importance is derived from lexical analysis of the LOs and is calculated against the existing teacher profile.

A careful analysis of the existing approaches highlights the fact that key teachers' individual professional capacity characteristics, such as their aforementioned ICT Competences, are not being currently exploited for providing personalized recommendations to support their daily teaching practice of course design and delivery. Incorporating such information can lead to more focused recommendations by forming better teacher neighborhoods and by filtering candidate LOs to identify the most appropriate for each individual teacher's ICT competences. Our previous work [15] has presented initial evidence that incorporating this type of teacher characteristics in the LO recommendation process depicted in a formal manner (i.e., the UNESCO ICT Competency Framework for teachers [55]) and appropriately mapping them (e.g., [11]) to the specific LO metadata schema that each dataset employs (e.g., the open discovery (ODS) space Repository LO Metadata Application Profile [56]) can have a significant positive impact on the predictive accuracy of the RS.

Moreover, due to the fact that (a) competences are not reified characteristics, but evolve over time, and (b) that such data are rarely (much less correctly) provided by the teachers themselves, profiling mechanisms should strive to construct, and maintain up-to-date, highly granulated ICT Competence profiles for teachers based on their relevance feedback data [26].

In the light of the above, a research problem can be identified, namely how can a teacher ICT Competence profile be (a) elicited based on the implicit and explicit relevance feedback data of the teachers and (b) utilized for delivering more informed LO recommendations. To the best of our knowledge, such an approach has not been previously considered in the field of RS. Such a system could provide a unified solution for providing personalized LO recommendations to individual teachers, since it would both automatically create and update teacher ICT competence profiles and exploit them towards informed LO suggestions.

The contribution of this paper, therefore, is the presentation of a proposed solution towards tackling this identified research problem. The first step, namely, teachers' ICT-CP elicitation, has already been analysed in our previous work [16]. In the context of this paper, however an additional evaluation of this step was performed in order to further enhance validattion of its robustness. The second step of the research problem, namely the exploitation of the generated ICT-CP for enhanced LO recommendations, is the original contribution of this paper.

3 ICT Competence-Based Learning Object Recommendations for Teachers

This section presents the proposed solution to the identified research problem, i.e., a new RS for providing enhanced LO recommendations to teachers based on their elicited ICT-CP. The proposed system is schematically depicted in Fig. 1.

It essentially comprises two main layers, as follows:

1. *Teacher ICT competence profile elicitation layer*. The first Layer, which is described in detail in Section 3.1, relates to the elicitation of the active teacher's ICT-CP

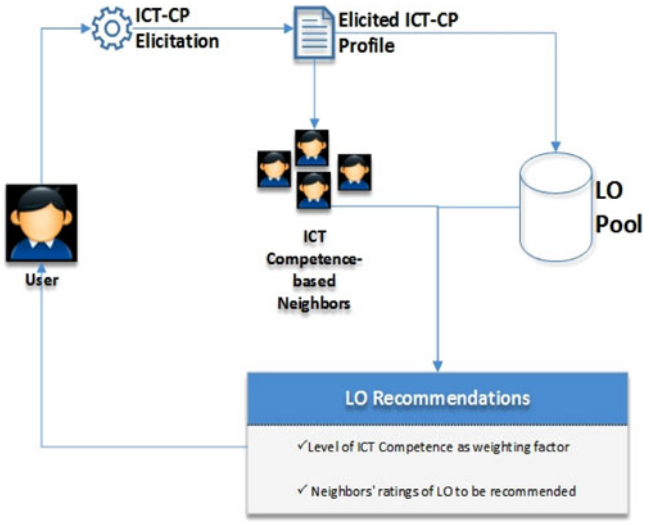


Fig. 1. Proposed recommender system overview.

based on their relevance feedback data. More specifically, the system creates a unified representation of the active teacher by harvesting his/her "Usage Data" from the LOR. Following that, this unified representation is fed to the ICT-CP Elicitation mechanism and the fuzzy ICT-CP of the teachers is generated.

2. *ICT competence-based learning object recommendation generation layer*. The second Layer of the proposed RS is the LO recommendation generation. This is performed by utilizing the output data generated from the first Layer. More specifically, as it will be further described in Section 3.2, it utilizes these output data in a dual manner, i.e., (a) for the selection of the active teacher's neighbors, in terms of their ICT-CP similarity to the active teacher and (b) for weighting each candidate LO in terms of its appropriateness for the active teacher's ICT-CP. This is a promising approach, which was introduced in our previous work [15] and was shown to result to more accurate LO recommendations.

The presentation of the RS will be performed in two steps in accordance to the two Layers of the RS, i.e., the teacher ICT Competence Profile Elicitation Layer and the ICT Competence-based LO Recommendation Generation Layer.

3.1 Teachers' ICT Competence Profile Elicitation Layer

3.1.1 Teacher Relevance Feedback Data

The first Layer of the proposed RS relates to the elicitation of the teachers' ICT Competence profiles based on their relevance feedback data from their interaction within any LOR.

Towards tackling this issue, a specific and appropriate set of relevance feedback data had to be selected. More specifically, this set should provide a valid proxy of teachers' ICT competence over the the available LO type in the LOR. More specifically, the LO types referred to the distinct metadata attributes which are used to characterize LO in LORs. It should be noted that these attributes are flexible and can be adapted to meet the IEEE LOM Application Profile that each LOR adopts, in line with the official IANA MIME type extensions (http://tinyurl.com/ol2lv34).

A set of four types of relevance feedback data (from now on referred to as "Usage Data"), commonly used in the literature, was selected, as follows:

1. *Rating history*. This category included the ratings teachers had provided over the available LO. To accommodate the unique rating patterns of each teacher (e.g., some tend to rate high by default), all ratings were first normalized by subtracting the rating mean of the active teacher. Rating data could provide a solid proxy of ICT competence due to the fact that a *high* rating could imply that the user has either actually used the LO or, at least, is comfortable with it [5]. A low rating on the other hand, was not considered as an indicator of low competence, because a person would probably not rate an item low just because they would not be able to use it.
2. *Bookmarking history*. In the same vein as before, bookmarking a specific LO type repeatedly could provide insight on the fact that the active teacher utilizes this type of resources frequently in their daily teaching practice. Therefore, it could provide a solid proxy for inferring the level of relevant ICT competence of teachers.
3. *Learning object access history*. The access patterns of teachers could be exploited for capturing their ICT competences in the accessed LO types. More specifically, if specific LO types had been repeatedly accessed from a teacher, they could be regarded as commonly used by him/her. This could infer useful information about the teacher's ICT competence profile.
4. *Learning object creation history*. The last "Usage Data" category referred to active teachers' sharing history, i.e., the number of LOs that each teacher had uploaded to the LOR. These data could provide insight on the teachers' ICT Competence based on the assumption that a teacher, who had *created* (and shared) a specific LO type, would be competent in actually using them.

These "Usage Data" types were utilized by the system for eliciting the teachers' ICT Competence profiles. More specifically, for each teacher, this process included an initial harvesting of all LOs that any of the "Usage Data" were available for. This information was then fed to the ICT Competence Profile Elicitation mechanism, towards inferring the teachers' ICT-CP.

3.1.2. The Teacher ICT Competence Profile Elicitation mechanism

The ICT-CP Elicitation mechanism comprised two phases, namely the "Aggregation" Phase and the "Fuzzification" Phase (Fig. 2).

1. *The aggregation phase.* The aim of this phase is to create a unified representation of the teachers' interactions within the LOR. More specifically, for each teacher, all four "Usage Data" information per LO metadata attribute are merged in one "Interactions" metric. This aggregated metric provides an overview of the level of interaction that each teacher had with

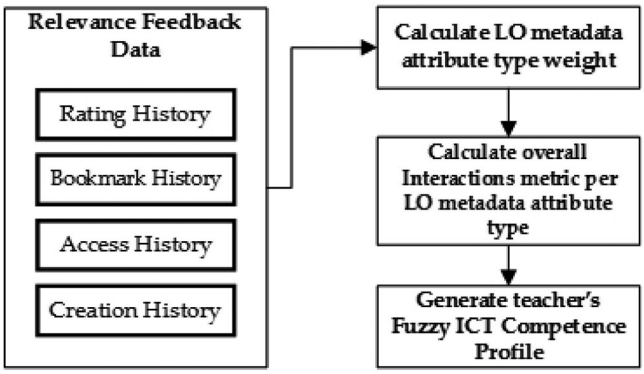


Fig. 2. Overview of the proposed RS Layer 1.

the diverse LO metadata types, by considering all four "Usage Data".

In order to calculate this metric, each LO metadata attribute type received a weight representing the level of significance that it had for each teacher, i.e., the level of preference that the active teacher demonstrated to this particular LO attribute type compared to the rest within the same "Usage Data" category.

The weights are calculated using the following formula for each teacher t:

$$w_i^t = \frac{\#LO_i}{\sum_{i=1}^{I} \#LO_i}, \tag{1}$$

where i refers to each metadata attribute type within the same "Usage Data" category and I refers to the overall cardinality of the metadata attribute set (which can vary between different LOR) within the same "Usage Data" category. The possible values for each weight range from [0 ... 1]. A weight value of "zero" for a specific LO metadata type signifies that the teacher has interacted with no LO of this type regarding the specific "Usage Data" category (e.g., no *ratings* provided for any LO of type "video"). On the other hand, a weight value of "one" for a specific LO metadata type signifies that the teacher has interacted solely with LO of this particular type regarding a specific "Usage Data" category (e.g., the teacher has provided *bookmarks* only for LO of type "video").

Moreover, (1) also normalizes the available "Usage Data", since the different "Usage Data" categories would probably not have the same cardinality of LO (i.e., there would probably be more "Access" data than "Bookmark" data).

Therefore, building on the weights calculated in (1) for each LO metadata type, the combined "Interactions" metric is computed using the following formula, for each teacher t:

$$Interactions_j^t = \frac{\sum_{j=1}^{J} w_j^t * \#LO_j}{\sum_{j=1}^{J} w_j^t}, \tag{2}$$

where, $Interactions_j^t$ is the resulting "Interactions" metric for each LO metadata attribute type j, J is the cardinality of the "Usage Data" set and w_j^t is the weight of the j LO metadata attribute, as calculated by (1).

$$\mu_{(\alpha)} = \begin{cases} 0, & if\ x_i \le x_{min}\ \textbf{or}\ x_i \ge x_{max} \\ \frac{x_i - x_{min}}{x_{max} - \frac{x_{min}+x_{max}}{2}}, & if\ x_{min} < x_i \le x_{min} + \frac{x_{min}+x_{max}}{2} \\ \frac{x_{max} - x_i}{x_{max} - \frac{x_{min}+x_{max}}{2}}, & if\ x_{min} + \frac{x_{min}+x_{max}}{2} < x_i < x_{max} \end{cases}$$

Fig. 3. Membership functions of the proposed RS.

Therefore, for each teacher t, the "Interactions" metric presents a unified and combinative depiction of their usage pattern in a LOR, for each LO metadata attribute type by considering all four "Usage Data". This is done by considering together (a) the significance that each LO metadata attribute type had to each teacher (i.e., the weights) and (b) the number of LO per metadata attribute type that the teacher had interacted with.

2. *The fuzzification phase*. After the completion of the Aggregation Phase process, and the construction of the active teacher's unified "Interactions" metric, the Fuzzification Phase is activated. This step is responsible for inferring the level of the teacher's ICT competence (their ICT-CP) by translating the aggregated "Interactions" metrics for each teacher to their fuzzy equivalents. The latter would be the final proxy of the teachers' ICT Competence. The reason for employing fuzzy logic is due to the fact that it is considered as an appropriate means of depicting data that are "vague" and difficult to assign to crisp categories [57]. Therefore, apart from the extensive use of this approach for creating user profiles in different application contexts [22], [58], fuzzy logic is selected due to the "fluid" nature of competences. More specifically, competences are not crisp characteristics, but can be attained in a continuum of proficiency levels and can be constantly updated [59].

The adopted fuzzification process uses a set of five linguistic variables depicting levels of competence, i.e., Very Low, Low, Medium, High and Very High. The use of five linguistic variables is a typical method in the literature (e.g., [29]). Additionaly, a triangular membership function was utilized, which is presented in Fig. 3. The latter was used due to the fact that it is a commonly used membership function type for depicting characteristics similar to ours [24], where $\mu_{(\alpha)}$ represents the membership function value and it is calculated for all five linguistic variables each time, x_i represents the "Interactions" metric value and the x_{min} and x_{max} depict the minimum and maximum marginal values for each Fuzzy linguistic variable. The membership function is schematically depicted in Fig. 4.

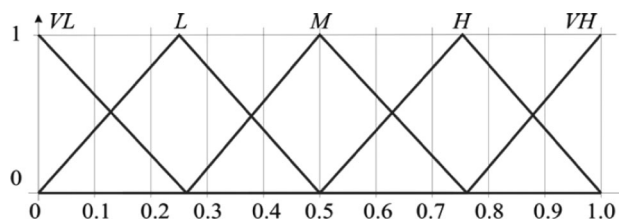


Fig. 4. Schematic Representation of the Membership function of the proposed RS.

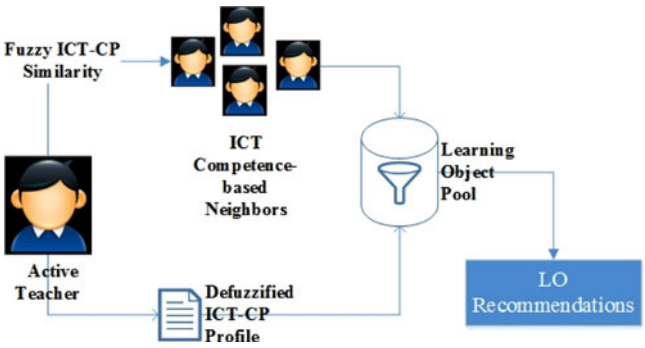


Fig. 5. Overview of the proposed RS Layer 2.

To give a brief example of the above, supposing that a teacher had a resulting aggregated "Interactions" value of "0.9" (from the Aggregation Phase of the ICT Competence Elicitation mechanism) regarding a specific type of LO (e.g., "inquiry-based" Lesson Plans), their generated fuzzy ICT-CP for this LO type would be:

VL	L	M	H	VH
0	0	0	0.4	0.6

More specifically, this means that the particular teacher is 40 percent Highly Competent in utilizing inquiry-based lesson plans and 60 percent Very Highly competent. Therefore, upon completion of the process of the Fuzzification Phase, each teacher's ICT-CP has been modeled in a fuzzy manner for all LO metadata types, based on his/her "Usage Data" within the LOR. The utilization of fuzzy logic for capturing and representing teachers' ICT Competences allows for more granulated construction of profiles. In turn, this enhanced level of teacher profile granularity can assist in providing more informed recommendations [22].

The exact manner in which this is realized in the proposed RS is described in the next section.

3.2 ICT Competence-Based Recommendations Layer

After the completion of the processes of the first Layer, the active teacher has been assigned with a unique fuzzy ICT Competence profile depicting their personal competences in each of the distinct LO metadata attribute types stored in the LOR. At this stage, the second Layer of the RS is activated, i.e., the LO recommendation generator (Fig. 5).

As aforementioned, the contribution in the proposed approach in this Layer related to the novel manner in which these recommendations were generated. More specifically, the elicited fuzzy ICT-CPs were utilized (a) as a means for alternative neighbor selection and (b) as a weighting factor for determining the appropriateness of each candidate LO to the active teacher's level of ICT competence.

The second Layer is divided in three phases, namely: (a) the Neighbor Selection Phase, (b) the ICT Competence Defuzzification Phase and (c) the Recommendation Generation Phase.

1. *The neighbor selection phase*: this phase aims at selecting the most suitable set of neighbors for the active teacher based on the similarity of their fuzzy ICT competence profiles instead of the commonly used rating similarity. In this way, the active teacher is provided with recommendations based on the opinions of colleagues that have the same ICT competence background with him/her, and thus (s)he is recommended with LO which will potentially be both useful and appropriate for them.The method employed for calculating this similarity is the Euclidean Distance, which is commonly used for similar processes [60].
2. *The defuzzification phase*: this phase aims to translate the unified fuzzy teacher ICT-CP in a single factor depicting the combined level of ICT competence of the teacher and incorporating all fuzzy levels. This factor will be utilized for weighting each candidate LO in terms of its suitability for the active teacher. The method employed for this Phase is the commonly used Center of Gravity method [61], which computes the center of gravity of the area under the membership function. The resulting factor of the Defuzzification Process is utilized in the second Phase of the ICT Competence-based Recommendations Layer of the RS.
3. *The recommendation generation phase*: this phase aims to generate and deliver the ICT Competence-based LO recommendations. As aforementioned, the proposed recommendation method utilizes the teachers' ICT-CP profiles in a dual manner.

First, the active teacher's neighbors are selected in Phase 1 ("Neighbor Selection Phase"), based on the similarity between their fuzzy ICT-CP. Second, the defuzzified ICT-CP factor is utilized as a filtering method of candidate LOs in terms of their appropriateness for the active teacher's level of competence.

Based on the above, for each LO i that is being considered for recommendation, the system assesses its appropriateness score (AS) for the active teacher t using the following formula:

$$AS_i^t = COG_{type\ of\ (i)} * \frac{\sum_{j=1}^{N} w_j * r_{ji}}{\sum_{j=1}^{N} w_j}, \quad (3)$$

where N is the amount of neighbors that had rated the specific LO, r_{ji} is the rating provided for LO i by neighbor teacher j and w_j is the Euclidean Distance between the active teacher and neighbor j. The Euclidean Distance is, therefore, utilized as a weighting factor in order to assign more gravity to the opinions of "closer" neighbors. Finally, the $COG_{type\ of\ (i)}$ is the output of the defuzzification process of the active teacher t. The resulting AS value represents the predicted rating for the active LO for the active teacher.

This section presented the proposed solution to the identified research problem, i.e., a unified approach for eliciting and exploiting teacher ICT-CP towards more informed LO recommendations. The next section presents the methodology employed and results generated for evaluating the performance of the proposed system, in terms of (a) the accuracy of the ICT-CP elicitation mechanism and (b) the predictive accuracy of the RS for generating LO recommendations.

4 Evaluation

4.1 Methodology

The evaluation methodology adopted in this paper follows the layered evaluation approach [17]. More specifically, monolithic evaluation of RS has been identified to lead to limited remedying potential in case of low RS accuracy. Essentially, this means that a potentially low accuracy of the RS cannot be linked to the specific element of the RS that is causing this level of performance. To alleviate this issue, the layered evaluation process proposes separate evaluation Phases for each of the RS Layers, namely Phase 1 refers to the evaluation of the Teacher Profiling Layer and Phase 2 refers to the evaluation of the ICT Competence-based Recommendation Layer. In this way, any underperforming Layer can be highlighted and focused remedying actions can be performed, towards increasing the accuracy of the specific Layer, and by extension, of the overall RS.

In the context of the present study, the Phase 1 evaluation focuses on the proposed system's accuracy in re-creating existing teacher ICT-CP. More specifically, the system generates a set of teacher ICT-CP based on their relevance feedback data. The generated ICT-CPs will then be contrasted against the existing ICT-CPs, which have already been provided voluntarily by the teachers themselves in the context of the ODS Project. Moreover, it should be mentioned that the existing ICT-CPs were provided by the teachers in a manual manner, using a webform and following the well-known UNESCO ICT Competency Profile for teachers.[1] The metric used for implementing this benchmark was the Jaccard co-efficient [62].

The Phase 1 evaluation has already been performed and reported in our previous work [16]. In the context of this paper, however, we retrieved an updated version of the same dataset and further validated the results, towards enhancing the robustness of the proposed method. The updated evaluation results of Phase 1, as well as a comparison to the initial evaluation results described in [16], are discussed in Section 5.1 of this paper.

Regarding the Phase 2 evaluation, the main focus was assessing the accuracy of the Layer 2 of the proposed approach to predict the ratings of the active teacher on LO they had not interacted with, using the process described in (3). This evaluation phase aims at providing evidence on the positive added value and the increased accuracy of the proposed RS (depicted as fuzzy hybrid, abbreviated as FH) compared to existing approaches. The commonly used root mean squared error (RMSE) metric was selected for measuring the predictive accuracy [63]. This metric is calculated based on the formula:

$$RMSE = \sqrt{\frac{1}{T}\sum_{u,i \in T}\left(\bar{r}_{ui} - r_{ui}\right)^2} \tag{4}$$

where $\bar{r}_{ui}$ is the set of generated predicted ratings for users u on items i, r_{ui} is the set of known ratings and T is the set of users and items for which the ratings are known. It should be noted that since this metric aims to capture errors in the predictions of the RS, lesser values of RMSE designate a better predictive accuracy of the RS.

As aforementioned, our proposed FH approach was benchmarked against a set of "control" recommendation methods, which are (a) commonly used in the literature and (b) have been reported to provide high levels of accuracy. More specifically, the control set included two types of recommendation approaches, namely user-based collaborative filtering (U) and item-based collaborative filtering (I) [62]. The former computes similarities between users to find the most similar users and predicts ratings based on how the item was rated by the most like-minded users. The latter approach shares the same idea but it is based on similarity between items rather than between users. For each of these two recommendation approaches one similarity measure was implemented, namely the Pearson correlation coefficient (PCC) for the user-based collaborative filtering [64] and the Adjusted Cosine correlation coefficient (CCC) for the item-based collaborative filtering [65]. Both similarity measures have been selected due to their reported high performance in terms of predictive accuracy in the context of their specific recommendation approach [64], [65].

The Pearson correlation coefficient is calculated based on the following formula:

$$PCC = \frac{\sum_{i \in I}(r_{ui} - \overline{r_u}) * (r_{wi} - \overline{r_w})}{\sqrt{\sum_{i \in I}(r_{ui} - \overline{r_u})^2 \sum_{i \in I}(r_{wi} - \overline{r_w})^2}}, \tag{5}$$

where I is the set of items that both users u and w have rated, r_{ui} and r_{wi} denote the ratings of the users u and w on the item i, while $\overline{r_u}$ and $\overline{r_w}$ denote the average ratings of the two users respectively.

The Adjusted Cosine correlation coefficient is calculated based on the following formula:

$$CCC = \frac{\sum_{u \in U}(r_{ui} - \overline{r_u}) * \left(r_{uj} - \overline{r_u}\right)}{\sqrt{\sum_{u \in U}(r_{ui} - \overline{r_u})^2}\sqrt{\sum_{u \in U}\left(r_{uj} - \overline{r_u}\right)^2}}, \tag{6}$$

where U is the set of users that have rated both items i and j, r_{ui} and r_{uj} denote the ratings of the user u on items i and j respectively while $\overline{r_u}$ denotes the average ratings of the user.

Therefore, the overall control recommendation method set consisted of two alternatives, i.e., user-based pearson correlation coefficient (UPCC) and the item-based adjusted cosine correlation coefficient (ICCC).

For all three cases (two control approaches plus the proposed approach), the predictive accuracy evaluation process was based on the standard "Leave-N-out" technique [66]. More specifically, this method includes splitting the available dataset in two sub-sets, namely the "training" set and the "test" set. The former contained 70 percent of the overall data and was used for training the RS and generating the recommendations. The latter contained the remaining 30 percent of the overall data and was used for evaluating the system accuracy, i.e., the recommendations generated from the training set [62]. Finally, the evaluation experiment was run for an increasing number of neighborhood size (one through twenty) in order to monitor the behavior of each approach in each occasion.

1. http://unesdoc.unesco.org/images/0021/002134/213475e.pdf [Accessed 10 May 2015]

4.2 Datasets

The present study utilized three educational datasets for supporting the two evaluation Phases, all of which were retrieved from existing learning object repositories which are currently in use. More specifically, the LORs used in our experiments were the ODS Repository http://portal.opendiscoveryspace.eu, the discover the cosmos (DtC) Repository http://portal.discoverthecosmos.eu, and the open science resources (OSR) Repository http://www.osrportal.eu/en/repository.

The ODS dataset, an earlier version of which was used in previous work [11], [15], [16], contained (a) existing, manually provided teachers' ICT Competence profiles, depicted using the UNESCO ICT Competency Framework for teachers (ICT-CFT) [55], (b) LO metadata records characterized using a specific IEEE LO Metadata Application Profile [56] and (c) "Usage Data" of the teachers as they were defined in Section 3.1.1. Due to the unique availability of existing teachers' ICT-CP, the ODS dataset was utilized for the Phase 1 evaluation process, for benchmarking the proposed teacher ICT Competence elicitation method. As aforementioned, this evaluation process was initially described in [16], where it was performed on an earlier instance of the ODS dataset. In the context of the work presented in this paper, an up-to-date version of the ODS dataset was retrieved towards further validating these initial promising evaluation results. In both experiments, towards the evaluation of the re-creation accuracy of the proposed method, mapping rules were utilized for connecting ICT Competences as they were described in ICT-CFT and LO metadata attributes as they were described in the IEEE LOM Application Profile of the ODS dataset [56]. These mapping rules were presented and evaluated in previous work [11].

The ODS dataset was not utilized in the Phase 2 evaluation process, in order to fully adhere to the layered evaluation framework adopted, namely to not only evaluate the different layers of the proposed RS individually, but also to utilize unique datasets for evaluating each layer.

The DtC dataset (http://portal.discoverthecosmos.eu) and the OSR dataset [67] contained (a) LO metadata records and (b) "Usage Data" of teachers that had been registered to the corresponding portals. The DtC and OSR datasets did not contain existing teacher ICT Competence profiles. Therefore, they were used only in the Phase 2 evaluation process, i.e., for evaluating the predictive accuracy of the proposed ICT Competence-based RS for teachers, by exploiting teachers' ICT-CP elicited by the first Layer of the proposed RS.

An overview of all three datasets is provided in the following Table 2. aggregation level (AL) 1 LOs refer to standalone Educational Resources (e.g., flash simulations, educational games, text documents), while aggregation level 2 LOs refer to Lesson Plans and/or Educational Scenarios (i.e., flows of learning activities supported by Educational Resources).

Overall Sample Size (N) refers to the total number of LOs in each AL category. The number of unique users refers to the cardinality of the set of teachers that had contributed at least in one "Usage Data" category. The remaing four data categories, namely Rating Data, Access Data, Creation Data and Bookmark Data, refer to the number of LOs that the corresponding "Usage Data" were provided for. As aforementioned, these four "Usage Data" categories were exploited by the proposed RS for eliciting the teachers' ICT-CP.

TABLE 2
Educational Datasets Overview

	ODS Dataset		DtC Dataset		OSR Dataset
	AL1	**AL2**	**AL1**	**AL2**	**AL1**
Overall Sample Size (N)	794	519	92,709	629	1,545
Rating Data	240	173	835	7,469	1,148
Access Data	3,308	392	183	708	5,026
Creation Data	794	519	92,709	629	1,545
Bookmark Data	177	45	47	39	345
Unique Users	686		209	281	829

The OSR dataset only contained usable data for Aggregation Level 1 Learning Objects. Finally, as "unique users" for the ODS dataset were only considered those who had contributed their ICT-CP, since the purpose of this dataset was to evaluate the first Layer of the proposed RS.

Furthermore, regarding the Phase 2 evaluation process, in order to allow for a more informed overview of the evaluation results (presented in Section 5), an analysis of specific characteristics of the datasets utilized (i.e., DtC AL1, DtC AL2 and OSR) was performed. More specifically, this analysis was performed due to the reported high dependence of all RS's performance to the dataset to which they are fed [62], [68] and the resulting need to robustly understand the shift in performance of the selected RS benchmark approaches between the different datasets. Based on the works of [68], [69], we selected a set of two dataset characteristics that have been reported to greatly influence the level of performance of RS. These are:

1. *Rating density*. It is defined as the ratio of known ratings against the number of all possible ratings that can be provided (i.e., $|\boldsymbol{Ratings}|/|U| * |I|$, where U and I are the number of users and items respectively). A logarithmic transformation was performed in order to normalize its values. Rating Density has been attributed with a high *positive* correlation to collaborative filtering RS performance, i.e., denser datasets can allow for more accurate recommendations [68].
2. *Rating standard deviation*. It is defined as the standard deviation of the ratings provided by the users in the dataset. Rating Standard Deviation has been attributed with a high *negative* correlation to collaborative filtering RS performance, i.e., lower levels of Rating Standard Deviation can lead to more accurate recommendations [68].

The instantiations of the abovementioned dataset characteristics for the selected datasets for the Phase 2 Evaluation are depicted in Table 3.

As Table 3 depicts, the densest dataset is the DtC AL2, followed by the OSR and the DtC AL1. Therefore, based on the conclusions of [68], we should expect that evaluation result accuracy in each of the three datasets should follow the same correspondingly descreasing order. This intuition is supported by the Rating Standard Deviation data, which

TABLE 3
Educational Dataset Characteristics

	DtC Dataset		OSR Dataset
	AL1	**AL2**	**AL1**
Rating Density	−4,365	−1,386	−3,047
Rating Standard Deviation	0,966	0,875	0,901

(given its negative correlation to RS accuracy) also signify the same correspondingly descreasing order in terms of expected evaluation accuracy.

Table 4 presents an overview of the evaluation methodology and datasets described in Sections 4.1 and 4.2.

The following section presents the results for Phase 1 and Phase 2 of the evaluation methodology described in this section.

5 Results

Following the layered evaluation approach, the results for each of the two Layers of the proposed RS are presented separately in the following sections.

5.1 Teacher ICT Competence Profile Elicitation Method Evaluation

As aforementioned, an initial evaluation of the Phase 1 was conducted in our previous work [16]. These preliminary evaluation results (Experiment #1) are presented in Fig. 6 along with the new Phase 1 evaluation results generated from the updated version of the ODS dataset (Experiment #2). Both sets of results provide evidence of the proposed method's high efficiency to elicit the ICT-CP of teachers based on their "Usage Data".

More specifically, for Educational Resources (AL1 LOs) the proposed system re-created the existing ICT-CP, which, as aforementioned, where provided by the teachers themselves in a manual manner through a web-form, with 70 percent accuracy in the initial Experiment #1 and with 73 percent accuracy in the updated Experiment #2. For Lesson Plans (AL2 LOs) the ICT-CP re-creation accuracy level was 81 percent in the initial evaluation Experiment #1 and 79 percent accuracy in the new evaluation Experiment #2.

Regarding the evaluation results' accuracy levels in their own regard, they are very promising for both evaluation experiments. The reported deviations from the original, user-provided ICT-CPs can be mainly attributed to the prominent reason that teachers could have provided incorrect descriptions of their ICT competences to begin with.

TABLE 4
Overview of Evaluation Methodology

RS Layer	Evaluation			
	Dataset	**Benchmark**	**Evaluation Focal Point**	**Evaluation Metric**
Layer 1	ODS	Existing teacher ICT-CP	ICT-CP Re-creation accuracy	Jaccard Coefficient
Layer 2	DtC, OSR	UPCC ICCC	Rating Predictive Accuracy	RMSE

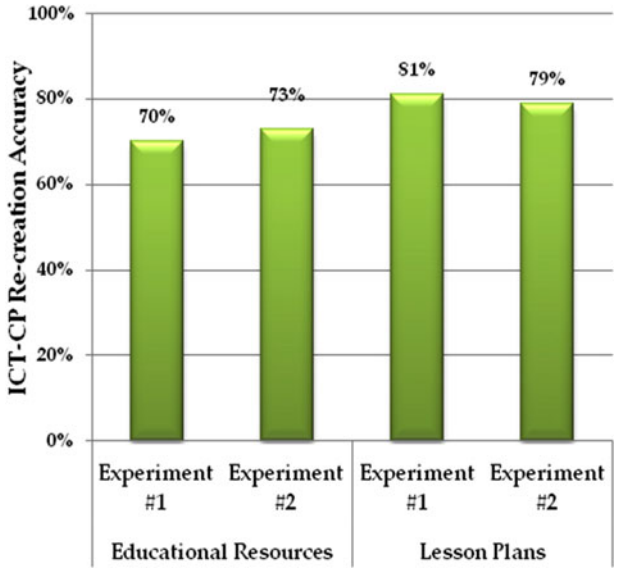


Fig. 6. Accuracy results for the two evaluation experiments of the Phase 1 of the teacher ICT Competence Elicitation Mechanism.

This is a very common issue in the user profiling literature, where users provide incomplete or false profiling data [21], [25]. Therefore, the manually-provided data would not always correctly represent the current level of ICT competence of the teacher that provided them, as opposed to the automatic elicitation method, which captures the teachers' *actual usage patterns* in using LOs and can therefore provide a more solid indicator of preference and competence [29]. Nonetheless, the re-creation accuracy results are significantly high in all experiments and for both Aggregation Levels despite this known shortcoming.

Moreover, regarding the result differences between Experiment #1 and Experiment #2, despite their small size, they could be attributed to the updated sets of "Usage Data" that the ODS dataset contained. These potential data alterations could have triggered these minor accuracy changes of the system, since the actual teachers' usage patterns could have been slightly altered since the initial version, without a corresponding manual change in the ICT-CP by the teacher. In any case, however, as Fig. 6 depicts, the evaluation result deviations between the two Experiments are very small and the absolute evaluation results are consistent between Experiments, indicating the proposed elicitation method's robustness.

Overall, the evaluation results for the first Layer of the proposed system were promising and provided with evidence that the teacher ICT-CP Elicitation Method could be utilized within learning object repositories, in order to elicit the teachers ICT-CP. As aforementioned, this information could allow for enhanced LO recommendations. The evaluation results for the latter, which represents the second Layer of the proposed system, are presented in the following section.

5.2 ICT Competence-Based Learning Object Recommendations Evaluation

The second Phase in the layered evaluation of the proposed approach to LO recommendations focused on the added value it could provide in terms of predictive accuracy, by

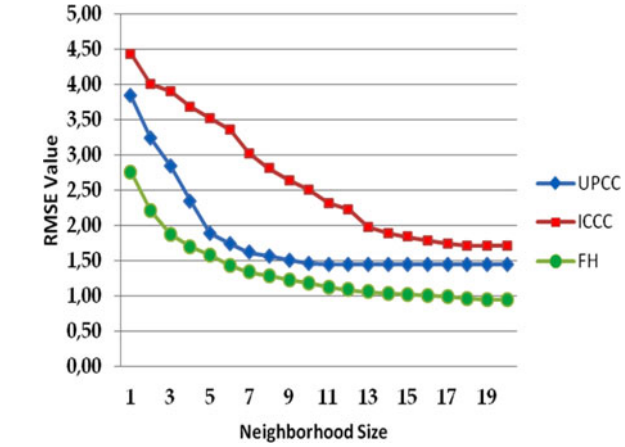


Fig. 7. Preditive accuracy evaluation results for AL1 Learning Objects in the DtC dataset.

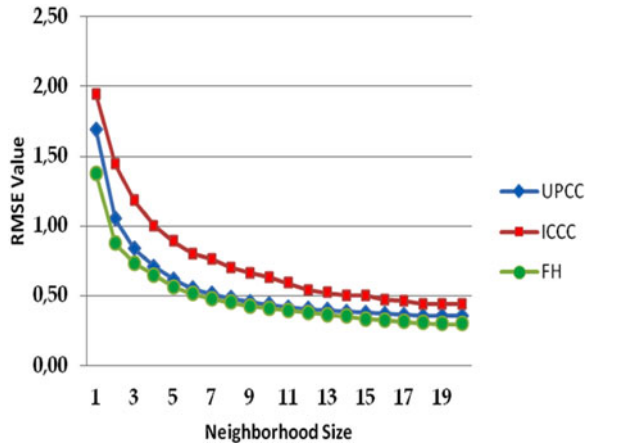


Fig. 8. Preditive accuracy evaluation results for AL2 Learning Objects in the DtC dataset.

utilizing the generated teacher ICT-CP from the first Layer. As aforementioned, the DtC (for AL1 and AL2 LO) and OSR (for AL1 LO) datasets were utilized in this process. The evaluation results for each dataset will be presented in separate.

1. *Discover the Cosmos dataset*. Figs. 7 and 8 depict the evaluation results of the DtC dataset for the two Aggregation Levels of learning objects that the dataset contained. As the Fig. 7 depicts, the proposed approach outperfoms both control recommendation methods for the AL1 LOs.

   Even though the accuracy of the two benchmark methods increases considerably as the neighborhood size (NS) grows, they are outperformed by the proposed FH approach for all NS values. Moreover, in the user-based control method, the threshold neighborhood size is around 10, meaning that increasing the neighbors above that threshold does not offer significant increase in accuracy.

   This is due to the fact that relatively few ratings and users were available in the DtC dataset for the AL1 LOs. Therefore, the user neighborhoods had limited data to be built on. This is closely related to the influence of dataset characteristics described in Section 4.2, and more specifically, the Rating Density [68]. For the ICCC method this occurs at a smaller degree, since, regarding candidate neighbors at least, there is a larger pool to choose from (i.e., the AL1 LOs of the dataset). The ICCC accuracy (as well as the proposed FH's accuracy), therefore, continues to improve for a larger NS span, and begins to plateu at around 15 neighbors. Therefore, the findings from this dataset conclude that the proposed FH approach provides more accurate LO recommendations, throughout the NS span.

   Regarding AL2 LOs (Fig. 8), the proposed approach similarly outperfoms both control recommendation methods for all NS values. Moreover, it is important to notice that all three methods' predictive accuracy is considerably increased, compared to their AL1 equivalents.

   The reason for this universal performance improvement can be attributed to the dataset characteristics of the DtC AL2 dataset and its comparison to its AL1 equivalent. More specifically, as Table 3 depicted, the DtC AL2 dataset was much denser (Rating Density) and had a lesser value of Rating Standard Deviation. Therefore, building on (and confirming) the findings of [68], we mainly attribute the difference of the results of Figs. 7 and 8 (and the enhanced predictive accuracy of the RS in the DtC AL2 dataset), to these two dataset characteristics' values.

   Moreover, the significant improvement of Rating Density (compared to the DtC AL1 dataset) increased the NS threshold that provided better predictive accuracy for all three recommendation methods. More specifically, the NS threshold for the UPCC control method was increased to around 17 and for the proposed FH, as well as for the ICCC control method, the NS threshold is around 18.

   Overall, despite the significant improvement in all three RS methods in the DtC AL2 datasets (compared to the DtC AL1), the proposed FH still outperformed the two benchmark methods and, essentially, delivered more accurate LO recommendations.

2. *Open science resources dataset*. As aforementioned, the OSR dataset only contained usable data for AL1 LOs. The accuracy evaluation results are presented in Fig. 9.

   As the Fig. 9 depicts, the proposed approach again ourperforms both benchmark methods for all neighborhood sizes.

   Moreover, the predictive accuracy of all approaches is decreased compared to the DtC AL2 dataset, but are better that the DtC AL1 dataset. This finding can be explained by considering the dataset characteristics presented in Table 3. More specifically, as the Table 3 data depict, the values for both OSR dataset charasteristics (i.e., Rating Density and Rating Standard Deviation) are better that the DtC AL1 and worse than the DtC AL2. Therefore, the predictive accuracy results depicted in Fig. 9 (and their connection to the results of Figs. 7 and 8) can be attributed to these facts. Regarding the NS thresholds, the results follow a similar pattern, i.e., they are increased compared to the DtC AL1 dataset and

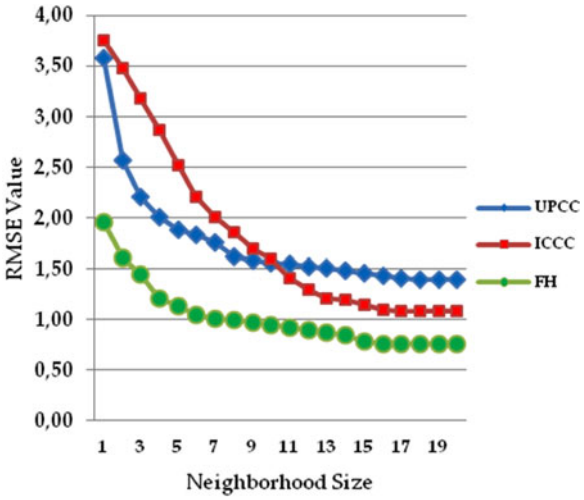


Fig. 9. Preditive accuracy evaluation results for AL1 Learning Objects in the OSR dataset.

decreased compared to the DtC AL2 dataset. More specifically, the UPCC control method plateus at around 16 neighbors, and the ICCC control method, as well as the proposed FH approach plateu at around 17 neighbors.

Therefore, the OSR dataset re-validated the promising results of the two DtC datasets, i.e., that the proposed FH approach can deliver more accurate LO recommendations to teachers, for all NS.

Overall, a consolidated overview of the predictive accuracy of the proposed FH approach provides with evidence of the added value that it can offer. More specifically, the reported promising results from Phase 2 evaluation experiments in the DtC (for both AL1 and AL2 LOs) and OSR datasets show that the proposed approach can deliver more accurate LO recommendations to teachers based on their elicited ICT Competence profiles. This is in line with the findings of our previous work [15] which presented initial evidence that incorporating (existing) teachers' ICT-CP in the LO recommendation process can provide with increased accuracy and, essentially, better results.

6 Conclusions and Future Work

This paper introduced and evaluated a new RS for providing LO recommendations to teachers based on their ICT-CP elicited from their relevance feedback data.

More specifically, the proposed system was divided in two Layers, namely (a) the teacher ICT Competence Profile Elicitation Layer, which was targeted at eliciting and constructing the teachers' ICT-CP based on their relevance feedback data, and (b) the ICT Competence-based LO Recommendation Layer, which utilized the elicited teacher ICT-CP from Layer 1 for providing more informed LO recommendations.

The RS evaluation followed the same layered approach and provided with evaluation results for each system Layer. Regarding the Layer 1 evaluation, the expreriment presented in this paper was the second one being conducted and the consistency between the two Experiments provides with additional evidence of the proposed elicitation's method robustness. Overall, both individual Layer evaluations, as well as in combination (as a holistic system), provided with evidence that the proposed approach can generate LO recommendations to teachers, which are personalized to their level of ICT competence. Moreover, the increased granularity offered by the fuzzy depiction of ICT competences was shown to have provided an added value in the overall RS performance.

Therefore, the evaluation results show that the proposed RS has the potential to be used for assisting teachers in their everyday tasks of course design, lesson planning as well as their implementation and delivery, by facilitating the process of selecting and retrieving appropriate LOs that match the teachers' ICT competence levels.

Future work in this research agenda could include the investigation in a granulated manner of the level of influence that the different levels of teachers' ICT competence can have on the predictive accuracy of the proposed system. Moreover, it would be interesting to investigate the potential of adaptive neighbor selection. More specifically, this strand will aim to extend the work presented in this paper which included selecting the active teacher neighbors based on their ICT-CP similarity. The next step will be to select neighbors in an adaptive manner depending on (a) the specific metadata attribute type of candidate LO to be recommended and (b) the specific ICT competences of the candidate neighbors in this particular LO type. By performing this dynamic neighborhood creation, the "gravity" of each neighbor in the recommendation process will be properly adjusted to reflect not only their similarity to the active teachers' ICT-CP, but also their "credibility" to recommend this particular LO metadata attribute type. Put simply, from the initial pool of ICT competence-based neighbors (proposed in the present paper), those that are more highly competent in the particular LO type (that is to be recommended) will have an increased "gravity" of opinion.

Finally, further research should be focused on incorporating additional contextual factors in the LO recommendation process for teachers, such as the ICT Competences of their school [70], [71]. The reason for this is that such contextual factors (e.g., school infrastructure and school leader ICT competences) have been repeatedly reported as significant in the level of uptake of ICT in the formal learning processes [72]. Therefore, accommodating them within the LO selection process would potentially extend the scope of the proposed RS and improve its capacity to facilitate teachers in selecting appropriate LOs for their unique context of work and, therefore, to optimize their instructional design and delivery efforts.

Acknowledgments

The work presented in this paper has been partially supported by (a) the "Open Discovery Space: A socially-powered and multilingual open learning infrastructure to boost the adoption of eLearning Resources" Project that is funded by the European Commission's CIP-ICT Policy Support Programme (Project Number: 297229) and (b) "Inspiring Science: Large Scale Experimentation Scenarios to Mainstream eLearning in Science, Mathematics and

Technology in Primary and Secondary Schools" Project that is funded by the European Commission's CIP-ICT Policy Support Programme (Project Number: 325123). Moreover, the authors would like to acknowledge the Discover the Cosmos and the Open Science Resources Projects for providing the corresponding educational datasets. Until September 2015, D. G. Sampson was with the Department of Digital Systems, University of Piraeus, Greece.

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